

Bernoulli Spectral Closure for Selected Minimizers

This note attacks the remaining gap more directly than the boundary-layer deficit route. The conclusion is that the attempted estimate

$$D_I(0) + D_H(0) \lesssim E(U) - E(B_1)$$

is the wrong target: it is false for general nearly spherical graphs. The correct hard mechanism is the selected minimizer's Bernoulli law. Once the selected minimizer is in the global graph regime, the Euler-Lagrange condition itself suppresses every non-neutral spherical harmonic mode when the selection parameter σ is small.

This would give a direct spectral closure of the selected-minimizer contradiction if the tame Bernoulli remainder estimates stated below were proved. The follow-up Bernoulli expansion note corrects an invalid attempt at those estimates and records the route as conditional.

1 Why Boundary-Deficit Propagation Is Too Strong

Let

$$\partial\Omega_\varepsilon = \{(1 + \varepsilon\varphi(\theta))\theta : \theta \in \partial B_1\},$$

with φ a spherical harmonic of degree $k \geq 2$, normalized by $\|\varphi\|_{L^2(\partial B_1)} = 1$. The second variation of torsion at the ball has the $H^{1/2}$ scale

$$E(\Omega_\varepsilon) - E(B_1) \simeq \varepsilon^2 k.$$

On the other hand, the perimeter/isoperimetric boundary defect has the H^1 scale

$$D_I(0) \simeq \varepsilon^2 k^2.$$

Similarly, the first variation of the boundary Bernoulli datum contains the multiplier $k - 1$, so the Holder/Serrin defect satisfies

$$D_H(0) \simeq \varepsilon^2 (k - 1)^2.$$

Thus neither $D_I(0)$ nor $D_H(0)$ can be bounded linearly by the torsion energy on the full nearly spherical class. High frequencies violate the bound. The graph regime alone does not repair this, since small-amplitude high frequencies can keep $C^{1,\gamma}$ norms bounded while making the ratio D_I/E large.

Therefore the boundary-layer route can close sharply only by using the selected Euler-Lagrange equation, not by a general trace inequality.

2 Linearized Torsion Data at the Ball

Write the selected boundary as

$$\partial U_g = \{(1 + g(\theta))\theta : \theta \in \partial B_1\},$$

with

$$\int_{\partial B_1} g = O(\|g\|_{L^2}^2), \quad \int_{\partial B_1} \theta_i g = O(\|g\|_{L^2}^2),$$

coming from volume and barycenter normalization.

For the actual selected minimizer U_* , the volume is only $|B_1| + O(E(U_*) - E(B_1))$. This does not change the argument: first write ∂U_* as a graph over ∂B_{r_*} , where $|B_{r_*}| = |U_*|$, then dilate by r_*^{-1} . The Bernoulli law is preserved up to changing the constants Λ and a_σ , and $|r_* - 1| = O(E(U_*) - E(B_1))$ contributes only a quadratic/constant-mode error.

Let

$$u_g = u_{B_1} + u_1 + O(g^2), \quad u_{B_1}(r) = \frac{1 - r^2}{2N}.$$

The first shape derivative satisfies

$$\Delta u_1 = 0 \quad \text{in } B_1, \quad u_1|_{\partial B_1} = \frac{g}{N}.$$

If

$$g = \sum_{k \geq 0} g_k$$

is the spherical-harmonic decomposition, then

$$u_1(r, \theta) = \frac{1}{N} \sum_{k \geq 0} r^k g_k(\theta).$$

The outward normal derivative on the perturbed boundary is

$$\partial_\nu u_g = -\frac{1}{N} + \frac{1}{N} \sum_{k \geq 0} (k-1)g_k + O(\|g\|_{C^2}^2).$$

Equivalently, for the positive Bernoulli datum $q_g = |\nabla u_g|$,

$$q_g = \frac{1}{N} - \frac{1}{N} \sum_{k \geq 0} (k-1)g_k + O(\|g\|_{C^2}^2).$$

Thus the non-neutral linear operator is the Steklov operator shifted by one:

$$\mathcal{L}g := \sum_{k \geq 0} (k-1)g_k.$$

The kernel consists exactly of the $k = 1$ translation modes, while the volume constraint removes the $k = 0$ mode.

3 Linearization of the Selected Bernoulli Law

BDV Lemma 4.16 gives, on ∂U_g ,

$$q_g(x)^2 = \Lambda + a_\sigma \left[|x - x_U| - \left(\int_{U_g} \frac{y - x_U}{|y - x_U|} dy \right) \cdot x \right],$$

where

$$|a_\sigma| \leq C\sigma.$$

After centering $x_U = 0$, expand the bracket at the ball. On $x = (1 + g(\theta))\theta$,

$$|x| = 1 + g(\theta) + O(g^2),$$

and

$$\int_{U_g} \frac{y}{|y|} dy = \int_{\partial B_1} g(\theta) \theta d\theta + O(\|g\|_{L^2}^2).$$

The second term is the degree-one projection of g . Therefore the nonconstant part of the right-hand side is

$$a_\sigma (I - \Pi_1)g + O(\sigma \|g\|_{C^1}^2),$$

where Π_1 is the projection onto first spherical harmonics. Constants are absorbed into Λ .

On the left-hand side,

$$q_g^2 = \frac{1}{N^2} - \frac{2}{N^2} \mathcal{L}g + O(\|g\|_{C^2}^2).$$

Removing constants and first harmonics gives the projected equation

$$\boxed{\mathcal{L}g_{\geq 2} = O(\sigma)g_{\geq 2} + O(\|g\|_{C^2}^2)}.$$

Here $g_{\geq 2}$ denotes the projection onto spherical harmonics of degree $k \geq 2$.

4 Spectral Gap Closure

On the $k \geq 2$ subspace,

$$\|\mathcal{L}h\|_{L^2} \geq \|h\|_{L^2}.$$

Hence the projected Bernoulli equation yields

$$\|g_{\geq 2}\|_{L^2} \leq C\sigma \|g_{\geq 2}\|_{L^2} + C\|g\|_{C^2} \|g\|_{L^2}.$$

If

$$\sigma \leq \sigma_* \quad \text{and} \quad \|g\|_{C^2} \leq \delta_*,$$

with $C\sigma_* + C\delta_* \leq 1/2$, then

$$g_{\geq 2} = 0.$$

The volume and barycenter constraints give

$$g_0 = O(\|g\|_{L^2}^2), \quad g_1 = O(\|g\|_{L^2}^2).$$

Thus

$$\|g\|_{L^2} \leq C\|g\|_{L^2}^2.$$

For $\|g\|_{L^2}$ small, this implies

$$g \equiv 0.$$

Thus, conditional on the tame remainder estimates in Section 5, in the small-graph regime and for $\sigma \leq \sigma_*$, the only selected minimizer satisfying the Euler-Lagrange equation is the ball.

5 Nonlinear Estimates Needed for a Proof

The previous sections become a proof once the following two estimates are recorded. They are standard consequences of Schauder estimates for the Dirichlet problem on $C^{2,\gamma}$ graph domains and Taylor expansion of the normal map.

Lemma 5.1: Bernoulli map expansion. For $\|g\|_{C^{2,\gamma}} \leq \delta$, let $q_g = |\nabla u_g|$ on ∂U_g , pulled back to ∂B_1 . Then

$$q_g - \frac{1}{N} + \frac{1}{N} \mathcal{L}g = R_q(g),$$

with

$$\|R_q(g)\|_{L^2(\partial B_1)} \leq C\|g\|_{C^{2,\gamma}}\|g\|_{L^2(\partial B_1)}.$$

Sketch: pull u_g back to B_1 by the radial diffeomorphism $r\theta \mapsto r(1 + g(\theta))\theta$. The pulled-back equation is a uniformly elliptic perturbation of $-\Delta u = 1$. Subtract the ball solution and the harmonic first variation. Schauder estimates give a $C^{1,\gamma}$, hence $L^2(\partial B_1)$, remainder quadratic in the graph size.

Lemma 5.2: α -variation bracket expansion. Let

$$\Psi_g(\theta) = |(1 + g(\theta))\theta| - \left(\int_{U_g} \frac{y}{|y|} dy \right) \cdot (1 + g(\theta))\theta.$$

Then, modulo constants,

$$\Psi_g = (I - \Pi_1)g + R_\alpha(g),$$

with

$$\|R_\alpha(g)\|_{L^2(\partial B_1)} \leq C\|g\|_{C^1}\|g\|_{L^2(\partial B_1)}.$$

Sketch: expand $|(1 + g)\theta| = 1 + g$ and

$$\int_{U_g} \frac{y}{|y|} dy = \frac{1}{N} \int_{\partial B_1} (1 + g)^N \theta d\theta = \int_{\partial B_1} g\theta d\theta + O(\|g\|_{L^2}\|g\|_{C^1}).$$

The linear term $(\int g\theta) \cdot \theta$ is exactly the degree-one projection, up to the harmless normalization of the $Y_{1,i}$ basis.

Combining Lemmas 5.1 and 5.2 with the selected Bernoulli law gives the projected nonlinear equation

$$\mathcal{L}g_{\geq 2} = a'_\sigma(I - \Pi_1)g_{\geq 2} + R(g),$$

where $|a'_\sigma| \leq C\sigma$ and

$$\|R(g)\|_{L^2} \leq C(\|g\|_{C^{2,\gamma}} + \sigma\|g\|_{C^1})\|g\|_{L^2}.$$

This is the precise inequality used in Section 4.

6 Consequence for the Selection Contradiction

Conditionally, the selection principle produces a selected minimizer U whose asymmetry stays comparable to the original bad set:

$$\alpha(U) \geq (1 - o(1))\alpha(\Omega_0) > 0.$$

The graph-entry theorem places U in the small $C^{2,\gamma}$ graph regime once $\alpha(U) \leq \alpha_{\text{graph}}$. If the tame Bernoulli expansion were available, the spectral closure above would say that for small enough σ , U must be the ball. This would contradict $\alpha(U) > 0$.

This is a direct replacement for the boundary-deficit propagation target. It uses the two facts that are special to selected minimizers:

1. the Bernoulli coefficient is not arbitrary; it is an $O(\sigma)$ perturbation of a constant plus the controlled α -variation;
2. the linearized Bernoulli map has a spectral gap on all modes except volume and translations.

7 What Remains

The argument above is conditional. The missing estimates are not yet proved:

1. upgrade graph entry to a uniform $C^{2,\gamma}$ graph. BDV Lemma 4.16 gives smooth q_u , and Theorem 4.18 plus Schauder bootstrapping supplies this once the $C^{1,\gamma}$ graph is obtained;
2. write Lemmas 5.1 and 5.2 in full detail with a fixed pullback map and explicit dependence on the $C^{2,\gamma}$ radius;
3. track the small volume-normalization error from U_* to the unit-volume dilation.

Until these estimates are proved, the reliable closure remains BDV's nearly spherical second variation. No external Serrin stability theorem is needed for the BDV route.